

A. SUPPLEMENTARY MATERIAL

A.1. Construction of the surrogate function

Before proceeding with the proof, we derive a quadratic surrogate function, as presented in the following lemma.

Lemma 4. For a matrix variable $\mathbf{X} \in \mathbb{C}^{M \times N}$ and a positive definite matrix variable $\mathbf{Y} \succ \mathbf{0} \in \mathbb{C}^{M \times M}$, the following inequality holds

$$\log \det (\mathbf{I} + \mathbf{X}^H \mathbf{Y}^{-1} \mathbf{X}) \geq -\text{tr} (\mathbf{X}^H \tilde{\mathbf{U}} \text{diag} \{\gamma_n\} \tilde{\mathbf{U}}^H \mathbf{X}) + 2\Re \left(\text{tr} \left(\mathbf{U} \text{diag} \{\beta_n\} \tilde{\mathbf{U}}^H \mathbf{X} \right) \right) - \text{tr} \left(\tilde{\mathbf{U}} \text{diag} \{\alpha_n\} \tilde{\mathbf{U}}^H \mathbf{Y} \right),$$

where

$$\alpha_n = \frac{1}{1 + \lambda_n}, \quad \beta_n = \frac{\log(1 + \lambda_n)}{\lambda_n}, \quad \gamma_n = \frac{\beta_n - \alpha_n}{\lambda_n},$$

for $n = 1, \dots, N$. Here, matrix $\tilde{\mathbf{U}} = \mathbf{Y}^{-1} \mathbf{X} \mathbf{U}$, and $\mathbf{U} \mathbf{\Lambda} \mathbf{U}^H$ is the eigendecomposition of matrix $\mathbf{X}^H \mathbf{Y}^{-1} \mathbf{X}$, with $\mathbf{U}$ being a unitary matrix and $\mathbf{\Lambda} = \text{diag} \{\lambda_n\}$ a diagonal matrix of eigenvalues.

Proof. We first reformulate $f(\mathbf{X}, \mathbf{Y})$ as follows

$$\begin{aligned} \log \det (\mathbf{I} + \mathbf{X}^H \mathbf{Y}^{-1} \mathbf{X}) &\stackrel{(10.a)}{=} \int_0^1 \text{tr} \left(\left(\mathbf{I} + \tau \mathbf{X}^H \mathbf{Y}^{-1/2} \mathbf{Y}^{-1/2} \mathbf{X} \right)^{-1} \mathbf{X}^H \mathbf{Y}^{-1/2} \mathbf{Y}^{-1/2} \mathbf{X} \right) d\tau \\ &\stackrel{(10.b)}{=} \int_0^1 \text{tr} \left(\mathbf{X}^H \mathbf{Y}^{-1/2} \left(\mathbf{I} + \tau \mathbf{Y}^{-1/2} \mathbf{X} \mathbf{X}^H \mathbf{Y}^{-1/2} \right)^{-1} \mathbf{Y}^{-1/2} \mathbf{X} \right) d\tau = \int_0^1 \text{tr} \left(\mathbf{X}^H \left(\mathbf{Y} + \tau \mathbf{X} \mathbf{X}^H \right)^{-1} \mathbf{X} \right) d\tau \\ &\stackrel{(10.c)}{\geq} \int_0^1 \text{tr} \left(2\Re \left(\mathbf{X}^H \left(\mathbf{Y} + \tau \mathbf{X} \mathbf{X}^H \right)^{-1} \mathbf{X} \right) - \mathbf{X}^H \left(\mathbf{Y} + \tau \mathbf{X} \mathbf{X}^H \right)^{-1} \left(\mathbf{Y} + \tau \mathbf{X} \mathbf{X}^H \right) \left(\mathbf{Y} + \tau \mathbf{X} \mathbf{X}^H \right)^{-1} \mathbf{X} \right) d\tau \\ &\stackrel{(10.d)}{=} -\text{tr} \left(\mathbf{X}^H \tilde{\mathbf{U}} \mathbf{\Lambda}^{-1} \text{diag} \left\{ \int_0^1 \frac{\lambda_n \tau}{1 + \lambda_n \tau} \left(1 - \frac{\lambda_n \tau}{1 + \lambda_n \tau} \right) d\tau \right\} \tilde{\mathbf{U}}^H \mathbf{X} \right) \\ &\quad + 2\Re \left(\text{tr} \left(\mathbf{U} \text{diag} \left\{ \int_0^1 1 - \frac{\lambda_n \tau}{1 + \lambda_n \tau} d\tau \right\} \tilde{\mathbf{U}}^H \mathbf{X} \right) \right) - \text{tr} \left(\tilde{\mathbf{U}} \text{diag} \left\{ \int_0^1 \left(1 - \frac{\lambda_n \tau}{1 + \lambda_n \tau} \right)^2 d\tau \right\} \tilde{\mathbf{U}}^H \mathbf{Y} \right) \\ &\stackrel{(10.e)}{=} \text{tr} \left(\mathbf{X}^H \tilde{\mathbf{U}} \text{diag} \{\gamma_n\} \tilde{\mathbf{U}}^H \mathbf{X} \right) + 2\Re \left(\text{tr} \left(\mathbf{U} \text{diag} \{\beta_n\} \tilde{\mathbf{U}}^H \mathbf{X} \right) \right) - \text{tr} \left(\tilde{\mathbf{U}} \text{diag} \{\alpha_n\} \tilde{\mathbf{U}}^H \mathbf{Y} \right), \end{aligned} \tag{10}$$

where the equality (10.a) follows from Lemma 1, and the equality (10.b) uses the matrix identity $(\mathbf{I} + \mathbf{A}\mathbf{B})^{-1} \mathbf{A} = \mathbf{A}(\mathbf{I} + \mathbf{B}\mathbf{A})^{-1}$. For any positive definite matrix $\mathbf{Z}$, the inequality

$$\text{tr} (\mathbf{X}^H \mathbf{Z}^{-1} \mathbf{X}) \geq \text{tr} \left(2\Re \left(\mathbf{X}^H \mathbf{Z}^{-1} \mathbf{X} \right) - \mathbf{X}^H \mathbf{Z}^{-1} \mathbf{Z} \mathbf{Z}^{-1} \mathbf{X} \right).$$

holds². Let $\mathbf{Z} = \mathbf{Y} + \tau \mathbf{X} \mathbf{X}^H$. Then, inequality (10.c) holds. By the matrix inversion lemma, we have

$$\mathbf{Z}^{-1} = \mathbf{Y}^{-1} - \mathbf{Y}^{-1} \mathbf{X} \left(\frac{1}{\tau} \mathbf{I} + \mathbf{X}^H \mathbf{Y}^{-1} \mathbf{X} \right)^{-1} \mathbf{X}^H \mathbf{Y}^{-1}.$$

Let $\mathbf{X}^H \mathbf{Y}^{-1} \mathbf{X} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H$ be the eigenvalue decomposition of matrix $\mathbf{X}^H \mathbf{Y}^{-1} \mathbf{X}$, and $\tilde{\mathbf{U}} = \mathbf{Y}^{-1} \mathbf{X} \mathbf{U}$. Then,

$$\mathbf{Z}^{-1} = \mathbf{Y}^{-1} - \tau \tilde{\mathbf{U}} (\mathbf{I} + \tau \mathbf{\Lambda})^{-1} \tilde{\mathbf{U}}^H \tag{11}$$

Substituting this expression yields equality (10.d). Since the integrals evaluate to

$$\begin{aligned} \int_0^1 \frac{\lambda_n \tau}{1 + \lambda_n \tau} d\tau &= 1 - \frac{\log(1 + \lambda_n)}{\lambda_n}, \\ \int_0^1 \left(\frac{\lambda_n \tau}{1 + \lambda_n \tau} \right)^2 d\tau &= 1 + \frac{1}{1 + \lambda_n} - \frac{2 \log(1 + \lambda_n)}{\lambda_n}, \end{aligned}$$

we obtain equality (10.e), where $\alpha_n = (1 + \lambda_n)^{-1}$, $\beta_n = \lambda_n^{-1} \log(1 + \lambda_n)$, and $\gamma_n = \lambda_n^{-1} (\beta_n - \alpha_n)$. □

²This can be proved by expanding $\left\| \mathbf{Z}^{-1/2} \mathbf{X} + \mathbf{Z}^{1/2} \mathbf{Z}^{-1} \mathbf{X} \right\|_F^2 \geq 0$.

A.2. The proof of Theorem 3

Proof. Let $\mathbf{X} = \mathbf{H}_{i_k, k} \mathbf{W}_{i_k}$, $\mathbf{Y} = \mathbf{F}_{i_k}$, according to the Lemma 4, we have

$$\begin{aligned} \text{WSR}(\mathbf{W}) &= \sum_{(k, i)} \omega_{i_k} \log \det \left(\mathbf{I} + \mathbf{W}_{i_k}^H \mathbf{H}_{i_k, k}^H \mathbf{F}_{i_k}^{-1} \mathbf{H}_{i_k, k} \mathbf{W}_{i_k} \right) \\ &\geq \sum_{(k, i)} \omega_{i_k} \left[-\text{tr} \left(\mathbf{W}_{i_k}^H \mathbf{H}_{i_k, k}^H \mathbf{F}_{i_k}^{-1} \mathbf{H}_{i_k, k} \mathbf{W}_{i_k} \mathbf{U}_{i_k} \text{diag} \left\{ \gamma_{i_k, n_{i_k}} \right\} \mathbf{U}_{i_k}^H \mathbf{W}_{i_k}^H \mathbf{H}_{i_k, k}^H \mathbf{F}_{i_k}^{-1} \mathbf{H}_{i_k, k} \mathbf{W}_{i_k} \right) \right. \\ &\quad \left. + 2\Re \left(\text{tr} \left(\mathbf{U}_{i_k} \text{diag} \left\{ \beta_{i_k, n_{i_k}} \right\} \mathbf{U}_{i_k}^H \mathbf{W}_{i_k}^H \mathbf{H}_{i_k, k}^H \mathbf{F}_{i_k}^{-1} \mathbf{H}_{i_k, k} \mathbf{W}_{i_k} \right) \right) \right. \\ &\quad \left. - \text{tr} \left(\mathbf{F}_{i_k}^{-1} \mathbf{H}_{i_k, k} \mathbf{W}_{i_k} \mathbf{U}_{i_k} \text{diag} \left\{ \alpha_{i_k, n_{i_k}} \right\} \mathbf{U}_{i_k}^H \mathbf{W}_{i_k}^H \mathbf{H}_{i_k, k}^H \mathbf{F}_{i_k}^{-1} \left(\sum_{(l, j) \neq (k, i)} \mathbf{H}_{i_k, j} \mathbf{W}_{j_l} \mathbf{W}_{j_l}^H \mathbf{H}_{i_k, j}^H \right) \right) \right]. \end{aligned} \quad (12)$$

where

$$\begin{aligned} \alpha_{i_k, n_{i_k}} &= \frac{1}{1 + \lambda_{i_k, n_{i_k}}}, \quad \beta_{i_k, n_{i_k}} = \frac{\log(1 + \lambda_{i_k, n_{i_k}})}{\lambda_{i_k, n_{i_k}}}, \\ \gamma_{i_k, n_{i_k}} &= \frac{\beta_{i_k, n_{i_k}} - \alpha_{i_k, n_{i_k}}}{\lambda_{i_k, n_{i_k}}}, \quad \forall n = 1, \dots, D_{i_k} \end{aligned} \quad (13)$$

for $i = 1, \dots, I_k$, $k = 1, \dots, K$, and $\mathbf{U}_{i_k} \mathbf{\Lambda}_{i_k} \mathbf{U}_{i_k}^H$ is the eigenvalue decomposition of $\mathbf{SINR}_{i_k}$. For $i = 1, \dots, I_k$, $j, k = 1, \dots, K$, denote matrix

$$\begin{aligned} \tilde{\mathbf{U}}_{i_k} &= \mathbf{F}_{i_k}^{-1} \mathbf{H}_{i_k, k} \mathbf{W}_{i_k} \mathbf{U}_{i_k}, \\ \hat{\mathbf{A}}_{i_k, k} &= \mathbf{H}_{i_k, k}^H \tilde{\mathbf{U}}_{i_k} \text{diag} \left\{ \gamma_{i_k, n_{i_k}} \right\} \tilde{\mathbf{U}}_{i_k}^H \mathbf{H}_{i_k, k}, \\ \hat{\mathbf{A}}_{i_k, l} &= \mathbf{H}_{i_k, l}^H \tilde{\mathbf{U}}_{i_k} \text{diag} \left\{ \alpha_{i_k, n_{i_k}} \right\} \tilde{\mathbf{U}}_{i_k}^H \mathbf{H}_{i_k, l}, \\ \hat{\mathbf{B}}_{i_k} &= \mathbf{H}_{i_k, k}^H \tilde{\mathbf{U}}_{i_k} \text{diag} \left\{ \beta_{i_k, n_{i_k}} \right\} \tilde{\mathbf{U}}_{i_k}^H, \end{aligned}$$

and we have the following quadratic function

$$\begin{aligned} \text{WSR}(\mathbf{W}) &\geq \sum_{(k, i)} \omega_{i_k} \text{tr} \left(2\Re \left(\hat{\mathbf{B}}_{i_k}^H \mathbf{W}_{i_k} \right) - \mathbf{W}_{i_k}^H \hat{\mathbf{A}}_{i_k, k} \mathbf{W}_{i_k} \right) - \sum_{(k, i)} \sum_{(l, j) \neq (k, i)} \omega_{i_k} \text{tr} \left(\mathbf{W}_{j_l}^H \hat{\mathbf{A}}_{i_k, j} \mathbf{W}_{j_l} \right) \\ &= \sum_{(k, i)} \text{tr} \left(2\Re \left(\mathbf{B}_{i_k}^H \mathbf{W}_{i_k} \right) - \mathbf{W}_{i_k}^H \mathbf{A}_k \mathbf{W}_{i_k} \right). \end{aligned}$$

where $\mathbf{A}_k = \sum_{(l, j)} \omega_{j_l} \hat{\mathbf{A}}_{i_k, j}$, $\mathbf{B}_{i_k} = \omega_{i_k} \hat{\mathbf{B}}_{i_k}$. According to the MM algorithm framework [10], we consider the following problem

$$\begin{aligned} \max_{\mathbf{W}} \quad & \sum_{(k, i)} \text{tr} \left(2\Re \left(\mathbf{B}_{i_k}^H \mathbf{W}_{i_k} \right) - \mathbf{W}_{i_k}^H \mathbf{A}_k \mathbf{W}_{i_k} \right) \\ \text{s.t.} \quad & \sum_{i=1}^{I_k} \text{tr} \left(\mathbf{W}_{i_k} \mathbf{W}_{i_k}^H \right) \leq P_k, k = 1, \dots, K. \end{aligned} \quad (14)$$

By introducing a negative sign and converting the objective from maximization to minimization, we observe that Problem (14) is equivalent to Problem (8). Therefore, the proposed XMMSE algorithm adheres to the MM framework, and its convergence properties follow from the analysis in [10]. $\square$

Remark. For $k = 1, \dots, K$, $i = 1, \dots, I_k$, and $n = 1, \dots, D_{i_k}$, when the eigenvalue $\lambda_{i_k, n_{i_k}}$ approaches zero, direct application of equation (6) may lead to numerical instability. To mitigate this issue, we employ the Taylor series expansions

$$\beta_{i_k, n_{i_k}} = \sum_{t=0}^{\infty} (-1)^t \frac{\lambda_{i_k, n_{i_k}}^t}{t+1}, \quad \gamma_{i_k, n_{i_k}} = \sum_{t=0}^{\infty} (-1)^{t+1} \frac{t+1}{t+2} \lambda_{i_k, n_{i_k}}^t. \quad (15)$$

During the actual computation process, we truncate the series (15) to compute $\beta_{i_k, n_{i_k}}$ and $\gamma_{i_k, n_{i_k}}$ when $\lambda_{i_k, n_{i_k}} \leq \epsilon$, where $0 < \epsilon < 1$.